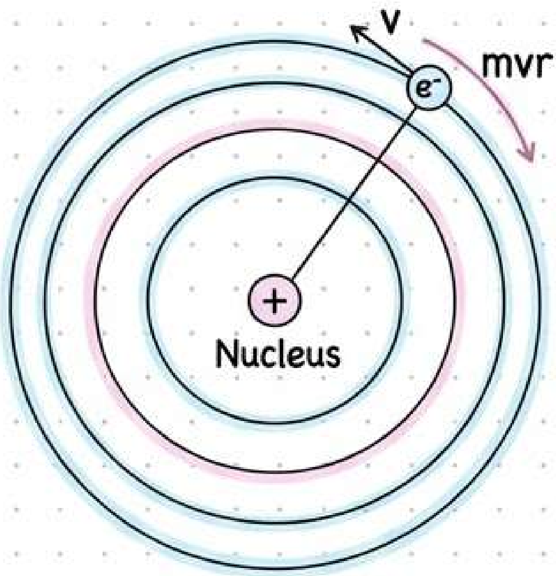
Atoms: Rutherford & Bohr Models



Rutherford Model:

Rutherford concluded that the majority of the space in an atom is empty. The entire positive charge and mass of an atom is concentrated in a small space known as the nucleus.
Distance of closest approach: Distance of a point from nucleus at which α-particles are nearest to the centre of nucleus.

Bohr's Model Postulates:

The centripetal force required for the circular motion of electrons is provided by the electrostatic force of attraction between the negatively charged electrons and the positively charged nucleus.

$$\frac{(Ze)e}{4\pi\varepsilon_0 r^2} = \frac{mv^2}{r}.$$

According to Bohr's second postulate, an electron can revolve only in certain fixed orbits known as stationary orbits. When an electron revolves in a stationary orbit its energy is constant and the angular momentum of the electron is an integral multiple of $h/2\pi$ where h is Planck's constant.

$$mvr = \frac{nh}{2\pi}.$$

According to Bohr's third postulate, when an electron jumps from an inner stationary orbit to an outer stationary orbit it absorbs energy and when it jumps back from an outer stationary orbit to an inner orbit it radiates energy.

$$E_i - E_f = h\nu = \frac{hc}{\lambda}.$$

Energy States:

Minimum energy required to free an electron from the ground state of an atom is called the ionization energy.

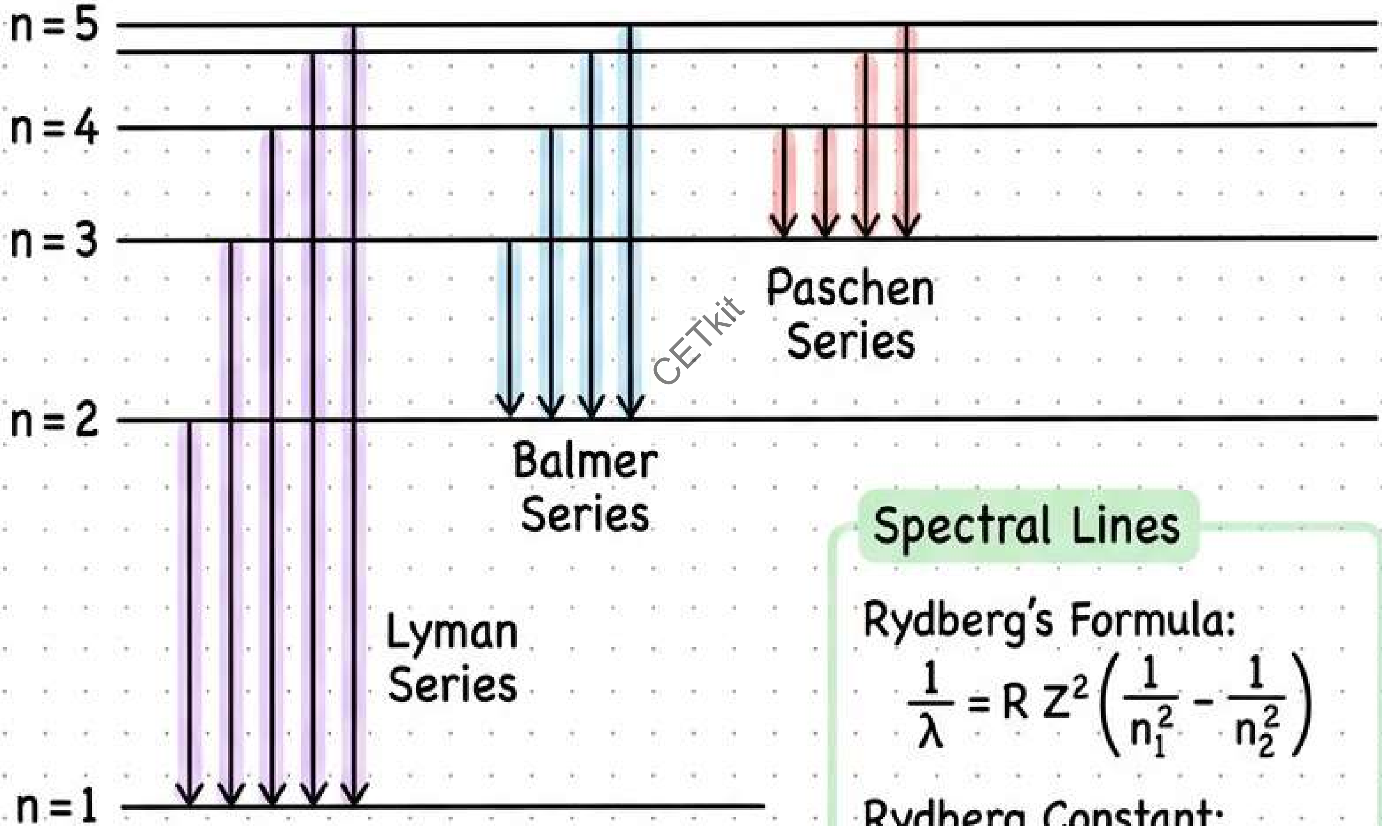
Hydrogen Spectrum & X-Rays

Hydrogen-Like Atoms

Radius of n^{th} orbit: $r_n = a_0\left(\frac{n^2}{Z}\right)$ $a_0 = 0.529$ Å

Energy in n^{th} orbit: $E_n = E_0\left(\frac{Z^2}{n^2}\right)$ $E_0 = -13.6$ eV

For attractive force $\propto 1/r^2$: TE = –KE = PE/2



Lyman (UV), Balmer (Visible), Paschen, Brackett, Pfund (IR)

Spectral Lines

Rydberg's Formula:

$$\frac{1}{\lambda} = R\,Z^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

Rydberg Constant:

$$R = 1.097 \times 10^7\ \text{m}^{-1}$$

X-Rays

Electromagnetic radiation of short wavelength (0.1 Å to 10 Å)

Cut off/Minimum Wavelength: $\lambda_{min} = \frac{12400}{V}$ Å

V = potential difference (in Volts)

Nuclei: Properties & Mass Defect

Characteristics of Nucleus

- The center of an atom, containing most mass.
- Density is nearly constant for all elements.
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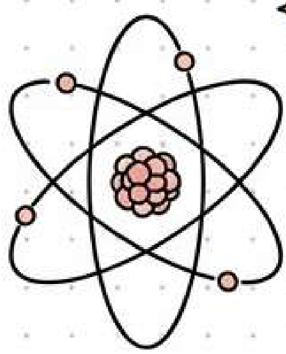
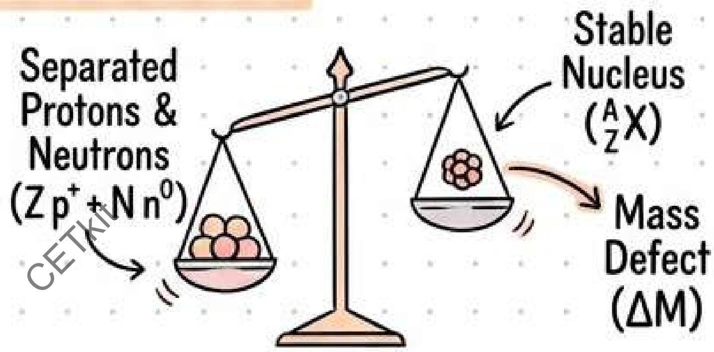
- Average radius : $R = R_0 A^{1/3}$ ($R_0 = 1.1 \times 10^{-15}$ m)



Mass-Energy & Binding Energy

Mass-Energy Equivalence

$$\boxed{E = mc^2}$$



$$\Delta M = (Zm_p + Nm_n) - M$$

Binding Energy (BE): Energy to split a nucleus into nucleons.

$$BE = (Zm_p + Nm_n - M)c^2$$

- High BE/nucleon = Very stable, takes more energy to split.

Q Value

Energy released in a nuclear reaction.

$$Q = (m_A + m_B - m_C)c^2 = BE_C - BE_A - BE_B$$

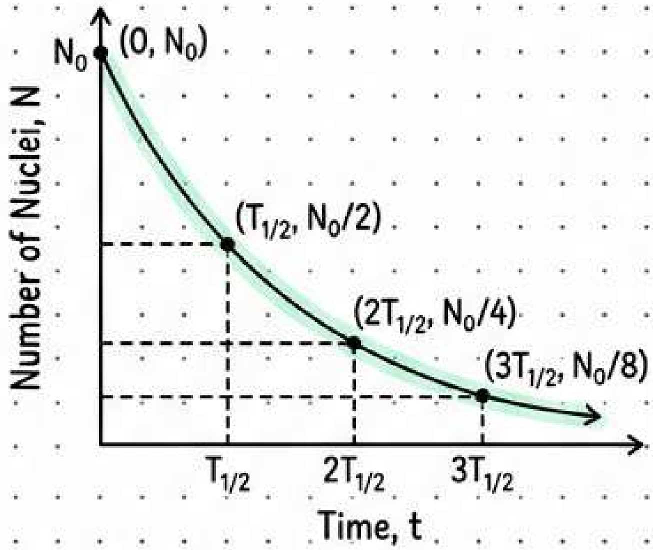
Radioactivity & Nuclear Reactions

Radioactive Decay Law

$$\text{Rate of decay} = \frac{dN}{dt} = -\lambda N$$

Rate of decay   Decay constant λ

$$N(t) = N_0 e^{-\lambda t}$$



Half-life & Average Life

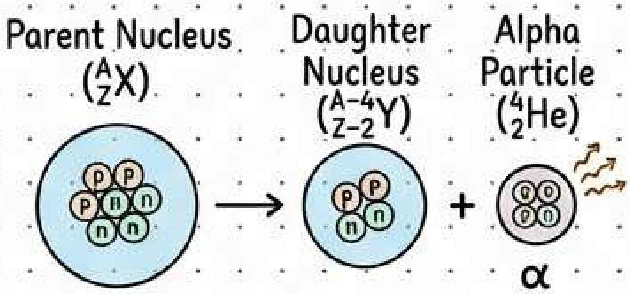
Half-life: $T_{1/2} = \frac{\ln 2}{\lambda} \approx \frac{0.693}{\lambda}$

Average life: $\tau = \frac{1}{\lambda} = \frac{T_{1/2}}{\ln 2} \approx 1.44\, T_{1/2}$

Activity of Sample

Activity: $R = \frac{-dN}{dt} = \lambda N = R_0 e^{-\lambda t}$ Becquerel (Bq)

Initial Activity R_0



$${}^{A}_{Z}X \longrightarrow {}^{A-4}_{Z-2}Y + {}^{4}_{2}He$$

Radioactive Decay Types

α **decay:** ${}^{4}_{2}He$ emission (slow, heavy)

β^- **decay:** Electron emission ($n \rightarrow p + e^- + \bar{\nu}_e$)

γ **decay:** High energy photon release from excited nucleus